

Mohammad Shahriar Hossain

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RESEARCH INTERESTS

Natural language processing and large language models, with a focus on contextual and semantic understanding, emotion and affective language understanding, figurative language, and polarization in text. Particularly interested in evaluating how LLMs interpret complex and implicit linguistic phenomena and in developing explainable methods for understanding model reasoning, behavior, and limitations.

EDUCATION

University of Alberta <i>M.Sc. Computing Science (Thesis)</i> Supervisor: Dr. Osmar R. Zaiane	Edmonton, AB Sep 2026 – Present
University of Alberta <i>B.Sc. Computing Science (Specialization in Software Practice)</i>	Edmonton, AB Sep 2021 – Jun 2026

WORK EXPERIENCE

Graduate Research Assistant <i>University of Alberta, Department of Computing Science</i> - Conduct research in natural language processing and large language models under the supervision of Dr. Osmar R. Zaiane, with a focus on language understanding, explainability, and reliable AI systems. - Contribute to the MIRA chatbot research project through literature review, experimentation, implementation, and evaluation of NLP and LLM based approaches.	Sep 2026 – Present Edmonton, AB
Undergraduate Teaching Assistant <i>University of Alberta, Department of Computing Science</i> - Mentor student teams on Agile and Scrum, lead labs and workshops, and help plan and deliver software projects in a collaborative setting. - Review Git/GitHub work with code reviews, testing, and peer feedback, and support clear communication through stand ups, sprint reviews, and client meetings.	Sep 2025 – Present Edmonton, AB
Data Analytics Co-op <i>Nutrien</i> - Analyze and visualize health, safety, procurement, and fleet data using Excel, Power Query, Power BI, SQL, and SAP HANA to build reports and dashboards for decision makers. - Build time series and risk prediction models in Python on AWS SageMaker and work with cross functional teams to improve data pipelines, automate workflows, and align machine learning solutions with business needs.	Jan 2025 – Present Calgary, AB
Software Developer Co-op <i>SensiImage Technologies Ltd.</i> - Maintained and extended the company web application across frontend and backend, using cloud infrastructure and basic DevOps practices to keep the product stable and responsive. - Designed and refined the display module interface and used data cleaning, analysis, and simple predictive models to improve product performance and user experience.	May 2024 – Dec 2024 Edmonton, AB

RESEARCH EXPERIENCE

NLP & LLM Research with Dr. Osmar R. Zaiane <i>University of Alberta, Department of Computing Science</i> - Conduct research on natural language processing and large language models, with an emphasis on language understanding, explainability, and reliable AI systems. - Contribute to the MIRA chatbot research project through literature review, experimentation, implementation, and evaluation of NLP and LLM based approaches.	2026 – Present Edmonton, AB
Cross-Lingual Semantics Research with Dr. Greg Kondrak <i>University of Alberta, Department of Computing Science</i> - Built a cross-lingual sense projection pipeline between English and Bengali using word sense disambiguation, machine translation, and word alignment to create semantic resources for a low-resource language. - Analyzed pipeline errors and added Bengali-side WSD, bilingual dictionary filters, and LLM-based checks so that projected senses better matched how words were used in context.	Sep 2025 – Apr 2026 Edmonton, AB

- Contributed to SemEval 2026 Task 9 on multilingual online polarization by preparing multilingual datasets and training an ensemble of XLM-R and DeBERTa models to classify polarization level and type.

LLMs and Figurative Language Research with Dr. Alona Fyshe

Sep 2025 – Apr 2026

University of Alberta, Department of Computing Science

Edmonton, AB

- Compared a multilingual encoder-only BERT model and a multilingual LLM on figurative vs. literal meaning classification across English, Bengali, Punjabi, Malayalam, and Portuguese.
- Used word sense disambiguation models in the pipeline to evaluate whether adding sense information improved performance for both the BERT model and the LLM.

ACADEMIC AND PERSONAL PROJECTS

Sense Aware Multilingual Polarization Detection | *XLM R, DeBERTa, WSD*

- Built a two path ensemble with multilingual encoders and sense aware features for POLAR@SemEval 2026 polarization and type classification, improving Macro F1 over lexical and LLM prompting baselines.

Cross Lingual Sense Projection for Bengali | *WSD, MT, Alignment, LLM Filtering*

- Implemented an English to Bengali sense projection pipeline with WSD, machine translation, and alignment, plus staged filters such as target side WSD, a bilingual dictionary, and LLM checks to improve coverage and precision of Bengali sense annotations.

Idiom Idiomaticity Detection (SemEval 2022 Task 2) | *Multilingual encoder, LLMs*

- Fine tuned a multilingual encoder and designed short prompts for instruction tuned LLMs to classify idiomatic vs literal uses of multiword expressions in English and Portuguese, with a shared F1 based evaluation setup.

Edmonton Housing Data Analysis | *SQL, Python, Power BI*

- Developed a Power BI dashboard using SQL and Python cleaned Edmonton housing data to help newcomers filter rentals by price, rooms, size, and community.

Predictive Modeling of Traffic Accidents | *R, ggplot2, dplyr, randomForest*

- Modeled accident severity using weather, road type, and time of day features with KNN, LDA, and Random Forest to support data informed road safety and traffic planning.

Oxygen: Chronic Illness Management App | *React Native, TypeScript, Expo, AI*

- Co developed a cross platform app for chronic illness patients and caregivers with symptom and appointment tracking, a secure medical wallet, caregiver access, and AI assisted medication and diagnosis support.

CO-CURRICULAR & EXTRA-CURRICULAR ACTIVITIES

Communications Director, International Students' Association

Jul 2021 - Apr 2024

University of Alberta

Edmonton, AB

- Planned International Students' Day and helped lead advocacy campaigns against international tuition increases.

UAlberta International Volunteer

Mar 2022 - Present

University of Alberta

Edmonton, AB

- Supported international students during orientation and transition, guiding them to campus services, programs, and community spaces.

Resident Assistant

Sep 2021 - Apr 2022

Residence Services, University of Alberta

Edmonton, AB

- Promoted equity, diversity, and inclusion through weekly check ins, floor events, and community activities.
- Handled student concerns and safety issues such as lockouts, alcohol incidents, first aid, and late night support.

AWARDS & SCHOLARSHIPS

International Student Scholarship

International Student Scholarship Fund, University of Alberta

- Merit-based scholarship awarded to international undergraduate students demonstrating strong academic performance.

University of Alberta International Country Scholarship

University of Alberta

- Awarded to students on study permits entering the first year of an undergraduate degree with superior academic achievement; selection based on academic standing, with preference for students from University of Alberta target regions.

ACHIEVEMENTS & COMPETITIONS

- 1st Place – CMPUT 401 Hackathon:** Job application tracker platform Sep 2024
- Built a job application tracking platform in a short hackathon, using software process and product management practices.
- 2nd Place – natHACKS 2024:** EEG based EMDR therapy aid Nov 2024
- Co developed an EEG based aid for EMDR inspired therapy, exploring how neurotechnology and ML can support mental health.
- PCL x BTMC Case Competition:** Interdisciplinary solution design Oct 2024
- Worked in an interdisciplinary team to design and present a data informed solution to an industry case, combining technical, business, and communication skills.